

First name:

Last name:

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Statistical Modelling Exam 25/01/2024

Exercise 1

The data contained in the `cement` dataset represent the hardness (`hardness` variable) of 13 types of cement with different chemical compositions. Specifically, each type is obtained with varying proportions of aluminium (`aluminium` variable), silicate (`silicate` variable), calcium aluminoferrite (`aluminium_ferrite`), and silicate bic (`silicate_bic`). The interest is explaining how the hardness of cement depends on the proportions of chemicals.

A regression model was fitted for this purpose and produced the following result:

	Estimate	Std. Error	<i>t</i> statistic	Pr(> <i>t</i>)
(Intercept)	124.4809	26.7557	4.653	0.0016
aluminium	0.9739	??	3.435	0.0089
silicate	-0.1405	0.2891	-0.486	0.6400
aluminium_ferrite	-0.4974	0.2751	??	??
silicate_bic	??	0.3214	-2.481	0.0381

Error sum of squares	49.378
Total sum of squares	2715.763
R^2 coefficient	??

- Write the model formulation and assumptions.
- Complete the missing values in the table. For “Pr(> |*t*|)” of `aluminium_ferrite` provide an approximate value. What variables have a statistically significant effect?
- Test the statistical hypothesis corresponding to the statement “the covariates do not have an effect on the hardness of cement”.
- On a reduced model (“model B”) that includes only the variables `aluminium` and `silicate_bic` the error sum of squares is equal to $SSE_B = 74.762$. Perform an F test to compare this model with the complete model (“model A”) that includes all the covariates. Interpret the result: which model would you prefer?
- Obtain the coefficient R^2 of model B. Instead of performing the test in point (d), could you have simply compared the coefficient R^2 of the two models? Why?
- Figure 1 shows two plots regarding the complete model (model A). Explain what they represent and interpret them.

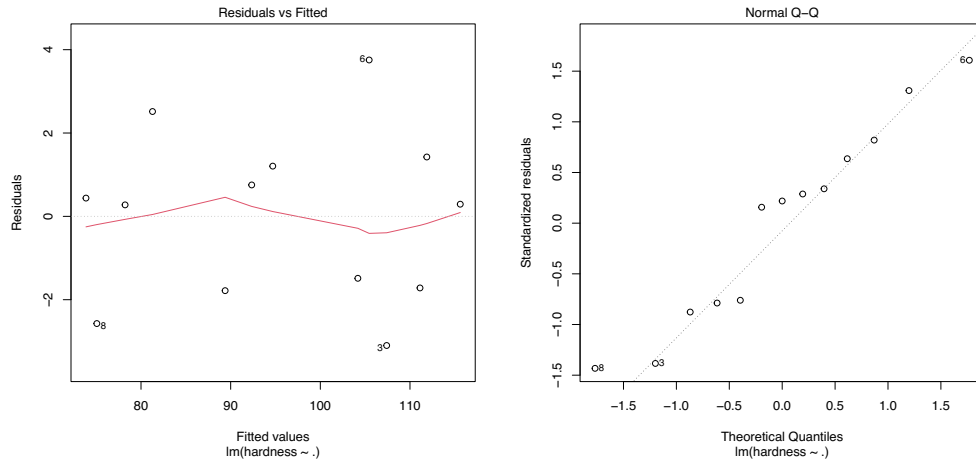


Figure 1:

Exercise 2

Let $(y_1, \dots, y_5)$ and $(y_6, \dots, y_{10})$ be two independent samples from a Poisson distribution of mean $\exp\{\beta_1\}$ and from a Poisson distribution of mean $\exp\{\beta_1 + \beta_2\}$, respectively.

- Formulate an appropriate Poisson regression model for the expected value of Y_i , $i = 1, \dots, 10$.
- Write the log-likelihood function of $\beta = (\beta_1, \beta_2)$ and the score function. Find the maximum likelihood estimate of (β_1, β_2) . Finally, obtain the observed information matrix.
- Determine an approximate distribution of the maximum likelihood estimator $\hat{\beta}$ of $\underline{\beta} = (\beta_1, \beta_2)$, and an approximate distribution of the maximum likelihood estimator $\hat{\beta}_1$ of β_1 .
- Provide the interpretation of the coefficient β_2 .
- Define the concept of “saturated model” and obtain the expression of maximum of the log-likelihood for this model.

		p						
		0.90	0.95	0.975	0.99	0.995	0.9975	0.999
standard Normal	z_p	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902
t with 4 df	$t_{4,p}$	1.5332	2.1318	2.7764	3.7469	4.6041	5.5976	7.1732
t with 5 df	$t_{5,p}$	1.4759	2.0150	2.5706	3.3649	4.0321	4.7733	5.8934
t with 6 df	$t_{6,p}$	1.4398	1.9432	2.4469	3.1427	3.7074	4.3168	5.2076
t with 7 df	$t_{7,p}$	1.4149	1.8946	2.3646	2.9980	3.4995	4.0293	4.7853
t with 8 df	$t_{8,p}$	1.3968	1.8595	2.3060	2.8965	3.3554	3.8325	4.5008
t with 9 df	$t_{9,p}$	1.3830	1.8331	2.2622	2.8214	3.2498	3.6897	4.2968
t with 10 df	$t_{10,p}$	1.3722	1.8125	2.2281	2.7638	3.1693	3.5814	4.1437
t with 11 df	$t_{11,p}$	1.3634	1.7959	2.2010	2.7181	3.1058	3.4966	4.0247
t with 12 df	$t_{12,p}$	1.3562	1.7823	2.1788	2.6810	3.0545	3.4284	3.9296
t with 13 df	$t_{13,p}$	1.3502	1.7709	2.1604	2.6503	3.0123	3.3725	3.8520

Table 1: Some quantiles of Gaussian and Student's t distribution: $p = \mathbb{P}(X \leq q_p)$. Columns correspond to probabilities p . Rows correspond to different distributions, in particular, for the t, each row corresponds to different degrees of freedom (df).

	0.90	0.95	0.975	0.99	0.995	0.9975	0.999
$f_{1,4;p}$	4.5448	7.7086	12.2179	21.1977	31.3328	45.6740	74.1373
$f_{1,5;p}$	4.0604	6.6079	10.0070	16.2582	22.7848	31.4067	47.1808
$f_{1,8;p}$	3.4579	5.3177	7.5709	11.2586	14.6882	18.7797	25.4148
$f_{1,13;p}$	3.1362	4.6672	6.4143	9.0738	11.3735	13.9468	17.8154
$f_{2,4;p}$	4.3246	6.9443	10.6491	18.0000	26.2843	38.0000	61.2456
$f_{2,5;p}$	3.7797	5.7861	8.4336	13.2739	18.3138	24.9640	37.1223
$f_{2,8;p}$	3.1131	4.4590	6.0595	8.6491	11.0424	13.8885	18.4937
$f_{2,13;p}$	2.7632	3.8056	4.9653	6.7010	8.1865	9.8392	12.3127
$f_{4,4;p}$	4.1072	6.3882	9.6045	15.9770	23.1545	33.3027	53.4358
$f_{4,5;p}$	3.5202	5.1922	7.3879	11.3919	15.5561	21.0478	31.0850
$f_{4,8;p}$	2.8064	3.8379	5.0526	7.0061	8.8051	10.9407	14.3916
$f_{4,13;p}$	2.4337	3.1791	3.9959	5.2053	6.2335	7.3728	9.0727
$f_{5,4;p}$	4.0506	6.2561	9.3645	15.5219	22.4564	32.2609	51.7116
$f_{5,5;p}$	3.4530	5.0503	7.1464	10.9670	14.9396	20.1783	29.7524
$f_{5,8;p}$	2.7264	3.6875	4.8173	6.6318	8.3018	10.2834	13.4847
$f_{5,13;p}$	2.3467	3.0254	3.7667	4.8616	5.7910	6.8200	8.3541
$f_{8,4;p}$	3.9549	6.0410	8.9796	14.7989	21.3520	30.6167	48.9962
$f_{8,5;p}$	3.3393	4.8183	6.7572	10.2893	13.9610	18.8022	27.6495
$f_{8,8;p}$	2.5893	3.4381	4.4333	6.0289	7.4959	9.2358	12.0455
$f_{8,13;p}$	2.1953	2.7669	3.3880	4.3021	5.0761	5.9318	7.2061
$f_{13,4;p}$	3.8859	5.8911	8.7150	14.3065	20.6027	29.5042	47.1627
$f_{13,5;p}$	3.2567	4.6552	6.4876	9.8248	13.2934	17.8667	26.2240
$f_{13,8;p}$	2.4876	3.2590	4.1622	5.6089	6.9384	8.5146	11.0596
$f_{13,13;p}$	2.0802	2.5769	3.1150	3.9052	4.5733	5.3113	6.4094

Table 2: Some quantiles of the F distribution: $p = \mathbb{P}(X \leq f_{df_1, df_2; p})$. Columns correspond to probabilities p . Rows correspond to different distributions, in particular, each row corresponds to different degrees of freedom (df).

EXERCISE 1

	Estimate	Std. Error	t statistic	Pr(> t)
β_1 (Intercept)	124.4809	26.7557	4.653	0.0016
β_2 aluminium	0.9739	0.2891	3.435	0.0089
β_3 silicate	-0.1405	0.2751	-0.486	0.6400
β_4 aluminium-ferrite	-0.4974	0.3214	-1.547	0.1381
β_5 silicate-bic	??	??	??	??

Error sum of squares	49.378
Total sum of squares	2715.763
R^2 coefficient	0.931

n=13

$$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 x_{i3} + \beta_5 x_{i4} + \epsilon_i$$

ALUMINUM SILICATE ALUMINUM-FERRITE
SILICATE-BIC

① $R^2 = \frac{SSR}{SST} = \frac{2666.385}{2715.763} = 0.981$

• $SSE = 49.378$

• $SST = 2715.763$

• $SSR = SST - SSE = 2715.763 - 49.378 = 2666.385$

② $\begin{cases} H_0: \beta_5 = 0 \\ H_1: \beta_5 \neq 0 \end{cases}$

$$t_{n-p} \stackrel{H_0}{\sim} \epsilon_{obs}^5 = \frac{\hat{\beta}_5 - \beta_5}{SE(\hat{\beta}_5)} \rightarrow \hat{\beta}_5 = \epsilon_{obs}^5 \cdot SE(\hat{\beta}_5) = -0.797$$

③ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

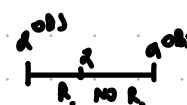
$$t_{n-p} \stackrel{H_0}{\sim} \epsilon_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} \rightarrow SE(\hat{\beta}_2) = \frac{\hat{\beta}_2}{\epsilon_{obs}^2} = 0.223$$

④ $\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases}$

$$t_{n-p} \stackrel{H_0}{\sim} \epsilon_{obs}^1 = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = -1.308$$

$$\alpha_{obs}^1 = P(|t_{n-p}| \geq |t_{obs}^1|) = 2 \min \{ P(t_{n-p} \leq -t_{obs}^1), P(t_{n-p} \geq t_{obs}^1) \} \approx 2(1-0.93) = 0.14$$

n-p=2



SINCE $\alpha_{obs}^1 > \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.01$; $\alpha = 0.05$

I DO NOT REJECT H_0

STATISTICALLY SIGNIFICANT: $\alpha_{obs}^1 < \alpha = 0.05$

• INTERCEPT

• ALUMINUM

• SILICATE-BIC

⑤

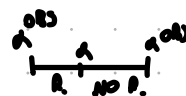
$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \\ H_1: \text{any} \end{cases}$$

• $SSE_1 = 49.378$

• $SSE_2 = SST = 2715.763$

$$F_{p-p_0; n-p} \stackrel{H_0}{\sim} f_{obs} = \frac{\frac{SSE_2 - SSE_1}{p - p_0}}{\frac{SSE_1}{n - p}} = \frac{\frac{2715.763 - 49.378}{5 - 1}}{\frac{49.378}{13 - 5}} = 107.993$$

$$\alpha_{obs}^2 = P(|F_{p-p_0; n-p}| \geq |f_{obs}|) \approx 1 - 0.999 = 0.001$$



SINCE $\alpha_{obs}^2 < \alpha$ I RES. $H_0 \rightarrow$ THE COVARIABLES HAVE AN EFFECT ON THE HARDNESS OF CEMENT

d

MOD A:

$$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \beta_4 x_{3i} + \beta_5 x_{4i} + \epsilon_i$$

ALUMINUM
SILICATE
ALUMINUM-FERIT
SILICATE-BIC

$$SSE_A = 49.373$$

$$SST = 2715.763$$

MOD B:

$$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_5 x_{4i} + \epsilon_i$$

ALUMINUM
SILICATE-BIC

$$SSE_B = 74.762$$

$$\begin{cases} H_0: \beta_3 = \beta_4 = 0 \\ H_1: \text{not } H_0 \end{cases}$$

$$F_{p-p_0, n-p} \stackrel{H_0}{\sim} F_{005} = \frac{\frac{SSE_B - SSE_A}{p-p_0}}{\frac{SSE_A}{n-p}} = \frac{\frac{74.762 - 49.373}{5-3}}{\frac{49.373}{13-5}} = 2.056$$

$$\alpha^{OBS} = P(F_{p-p_0, n-p} \geq F_{005}) \approx 1 - 0.83 = 0.17$$

↓
F_{2, 8}

α^{OBS} > α
R No R

since α^{OBS} > α for α = 0.1; α = 0.05; α = 0.01
I DO NOT REJECT H₀ → I WOULD PREFER THE
MODEL WITH JUST ALUMINUM AND SILICATE-BIC

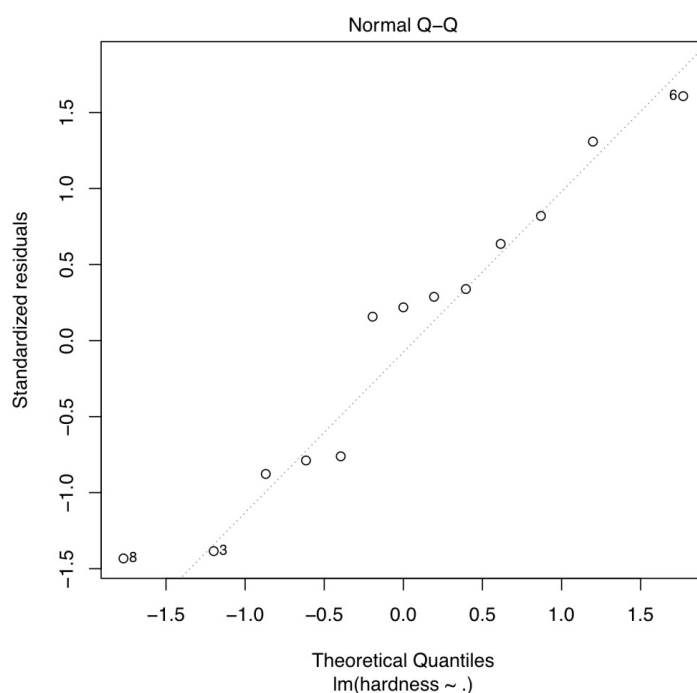
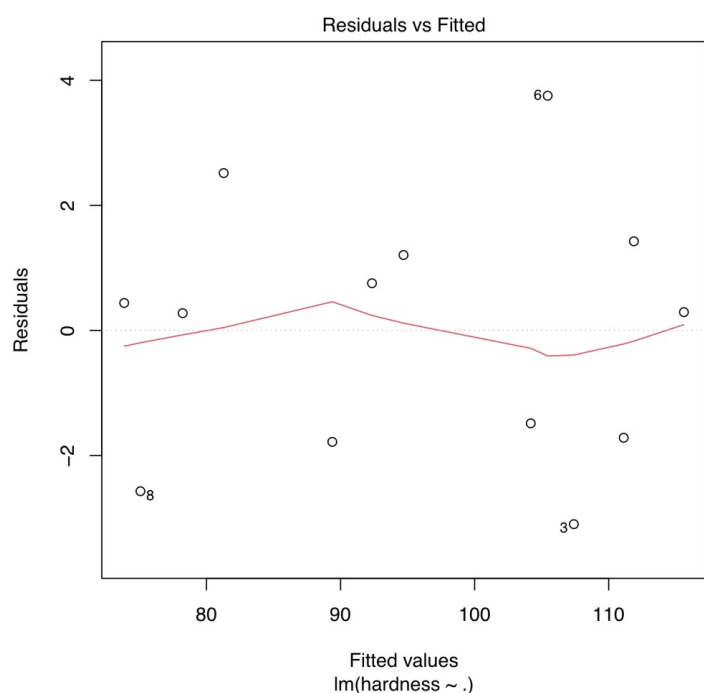
e

R²_B = ?

$$\begin{aligned} SST &= 2715.763 \\ SSE_B &= 74.762 \end{aligned} \rightarrow SSR_B = 2641 \quad R^2_B = \frac{SSR_B}{SST} = 0.972$$

NO. I COULD NOT COMPARE IN TERMS OF R²
SINCE IN NESTED MODELS SSE_B > SSE_A BY
DEFAULT, AND SO SSR_B < SSR_A SO, SINCE
SST IS CONSTANT IT IS OBVIOUS THAT R²_B WILL
BE LESS THAN R²_A

f



→ RESIDUAL PLOT SHOW THE
DISTRIBUTION OF RESIDUALS

- TOO FEW TO MAKE A GOOD ASSUMPTION (n=13)
- RANDOM CLOUD AROUND 0-LINE IS GOOD
- NO PATTERN/DISTRIBUTION IS VISIBLE (FOR NOW)

↓

- TOO FEW TO MAKE ASSUMPTIONS
- THERE ARE SOME LEFT AND RIGHT OUTLIERS AT THE TAILS
- IN GENERAL NOT CLOSE TO NORMAL DISTRIBUTION WHICH IS WHAT WE WANT IN THE ASSUMPTIONS
- ERROR SEEMS TO FOLLOW NORMAL DISTRIBUTION

EXERCISE 2

- $y_1, \dots, y_5 \quad Y_i \sim \text{Pois}(e^{\beta_1}) \quad i=1, \dots, 5 \quad N_i = e^{\tilde{x}_i^T \beta}$
 $y_6, \dots, y_{10} \quad Y_i \sim \text{Pois}(e^{\beta_1 + \beta_2}) \quad i=6, \dots, 10$

$$f(y_i; \mu_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \quad Y_i \sim \text{Poisson}(\mu_i) \quad i=1, \dots, 10$$

$$\mu_i = \beta_1 + \beta_2 x_i \quad x_i = \begin{cases} 0 & \text{for } i=1, \dots, 5 \\ 1 & \text{for } i=6, \dots, 10 \end{cases}$$

$$\log(\mu_i) = \eta_i$$

6 LIKELIHOOD $N_i = e^{\beta_1 + \beta_2 x_i} = e^{\tilde{x}_i^T \beta}$

$$L(N_i) = \prod \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}$$

LOG LIKELIHOOD

$$\ell(N_i) = \log(L(N_i)) = \log\left(\prod \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}\right) = \sum \log(e^{-\mu_i}) + \sum \log(\mu_i^{y_i}) - \sum \log(y_i!) = \sum -\mu_i + \sum y_i \log(\mu_i) - \sum \log(y_i!) = \sum -e^{\tilde{x}_i^T \beta} + \sum y_i \cdot \tilde{x}_i^T \beta - \sum \log(y_i!)$$

SCORE FUNCTION

$$s(N_i) = \begin{cases} a = \frac{d}{d\beta_1} \ell(N_i) \\ b = \frac{d}{d\beta_2} \ell(N_i) \end{cases}$$

$$\begin{cases} a = \frac{d}{d\beta_1} \left[\sum -e^{\tilde{x}_i^T \beta} + \sum y_i \cdot \tilde{x}_i^T \beta - \sum \log(y_i!) \right] = \frac{d}{d\beta_1} \left[\sum -e^{\beta_1 + \beta_2 x_i} + \sum y_i (\beta_1 + \beta_2 x_i) \right] = \sum -e^{\beta_1 + \beta_2 x_i} + \sum y_i \longrightarrow -\sum e^{\beta_1} - \sum e^{\beta_1 + \beta_2} + \sum y_i \\ b = \frac{d}{d\beta_2} \left[\sum -e^{\tilde{x}_i^T \beta} + \sum y_i \cdot \tilde{x}_i^T \beta - \sum \log(y_i!) \right] = \frac{d}{d\beta_2} \left[\sum -e^{\beta_1 + \beta_2 x_i} + \sum y_i (\beta_1 + \beta_2 x_i) \right] = \sum -e^{\beta_1 + \beta_2 x_i} x_i + \sum y_i x_i = \sum x_i (y_i - e^{\beta_1 + \beta_2 x_i}) \end{cases}$$

$$\downarrow$$

$$0 + \sum y_i - \sum e^{\beta_1 + \beta_2}$$

INFORMATION MATRIX

$$\begin{bmatrix} \frac{d s(N_i)_a}{d\beta_1} & \frac{d s(N_i)_a}{d\beta_2} \\ \frac{d s(N_i)_b}{d\beta_1} & \frac{d s(N_i)_b}{d\beta_2} \end{bmatrix} = \begin{bmatrix} \sum e^{\beta_1} - \sum e^{\beta_1 + \beta_2} & -\sum e^{\beta_1 + \beta_2} \\ -\sum e^{\beta_1 + \beta_2} & -\sum e^{\beta_1 + \beta_2} \end{bmatrix}$$

MLE:

$$\begin{cases} -\sum e^{\beta_1} - \sum e^{\beta_1 + \beta_2} + \sum y_i = 0 \longrightarrow e^{\beta_1} = \frac{\sum y_i - \sum e^{\beta_1 + \beta_2}}{\sum 1} \longrightarrow e^{\beta_1} = \frac{-\sum y_i + \sum y_i}{5} = \bar{y}_1 \\ \sum y_i - \sum e^{\beta_1 + \beta_2} = 0 \longrightarrow e^{\beta_1 + \beta_2} = \frac{\sum y_i}{5} = \bar{y}_2 \end{cases}$$

7 $\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \sim N_p \left(\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}; \mathcal{I}(\hat{\beta})^{-1} \right)$

• THE MARGINAL β_1 IS $\hat{\beta}_1 \sim N(\beta_1; \mathcal{I}(\hat{\beta})_{(1,1)}^{-1})$

• THE MARGINAL β_2 IS $\hat{\beta}_2 \sim N(\beta_2; \mathcal{I}(\hat{\beta})_{(2,2)}^{-1})$

8 $\begin{cases} \mu_1 = e^{\beta_1} \\ \mu_2 = e^{\beta_1 + \beta_2} \end{cases}$

$$\frac{\mu_2}{\mu_1} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2} \longrightarrow \beta_2 = \log\left(\frac{\mu_2}{\mu_1}\right) = \log\left(\frac{E(Y_i | \text{SAMPLE 2})}{E(Y_i | \text{SAMPLE 1})}\right)$$

• β_1 AND β_2 REPRESENT THE LOG OF THE RATIO BETWEEN THE MEANS OF THE TWO SAMPLES

SATVATED MODEL $D(\text{MODEL}) = 2 \{ \tilde{\lambda}(\text{SAT.}) - \hat{\lambda}(\text{MODEL}) \}$

SATVATED MODEL $\rightarrow$ model with $\# \text{ PARAM} = h$ (NUMBER OF OBSERVATIONS)